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Comparative Vertebrate Evolutionary Analyses of Type I Collagen: Potential of *COL1a1* Gene Structure and Intron Variation for Common Bone-Related Diseases

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Abstract

Collagen type I alpha 1 (*COL1a1*), which encodes the primary subunit of type I collagen, the main structural and most abundant protein in vertebrates, harbors hundreds of mutations linked to human diseases like osteoporosis and osteogenesis imperfecta. Previous studies have attempted to predict the phenotypic severity associated with type I collagen mutations, yet an evolutionary analysis that compares historical and recent selective pressures, including across noncoding regions, has never been conducted. Here, we use a comparative genomic and species evolutionary analysis representing ~450 My of vertebrate history to investigate functional constraints associated with both exons and introns of the >17-kb *COL1a1* gene. We find that although the COL1a1 amino acid sequence is highly conserved, there are both spatial and temporal signatures of varying selective constraint across protein domains. Furthermore, sites of high evolutionary constraint significantly correlate with the location of disease-associated mutations, the latter of which also cluster with respect to specific severity classes typically categorized in clinical studies. Finally, we find that *COL1a1* introns are significantly short in length with high GC content, patterns that are shared across highly diverged vertebrates, and which may be a signature of strong stabilizing selection for high *COL1a1* gene expression. In conclusion, although previous studies focused on *COL1a1* coding regions, the current results implicate introns as areas of high selective constraint and targets of bone-related phenotypic variation. From a broader perspective, our comparative evolutionary approach provides further resolution to models predicting mutations associated with bone-related function and disease severity.

Key words: skeleton, osteoporosis, vertebrate.

Introduction

Fibrillar collagens are the main structural proteins in vertebrates, abundant in connective tissue, cartilage, bone, and tendon (e.g., Gelse et al. 2003). In humans, fibrillar collagen proteins are linked to numerous skeletal diseases, such as osteoporosis, osteoarthritis, osteogenesis imperfecta (OI) and chondrodysplasia (e.g., Dalgleish 1997; Cohen 2006). The fibrillar collagen gene most commonly associated with disease is *COL1a1*, which encodes two of the three subunits of type I collagen (the third being encoded by *COL1a2*), is the most abundant protein in mammals, and is the main structural protein of bone, teeth, and tendon (Viguet-Carrin et al. 2006). Within this gene, >400 human disease-associated mutations (DAMs) have been identified, the majority of which are linked to osteoporosis afflicting ~200 million individuals globally, as well as OI (or brittle-bone disease) types I–IV and Ehlers–Danlos syndrome types I and VIIA (Dalgleish 1997; Marini et al. 2007) afflicting ~500,000 and 200,000, respectively (e.g., Stoll et al. 1989; Burrows 1999; Reginster and Burlet 2006). Interestingly, *COL1a1* is also associated with population variation in bone strength, measured as bone mineral density (Garcia-Giralt et al. 2002; Stewart et al. 2006; Jiang et al. 2007). Thus, as frequency differences among populations in osteoporosis are well documented (e.g., Lauderdale et al. 1997; Looker et al. 1997; Melton 1997; Barrett-Connor et al. 2005), *COL1a1* is a leading candidate gene in predicting not only severe bone-related diseases but also natural bone variation among populations in general.

The majority of known COL1a1 DAMs affect protein-coding regions, typically within the triple-helix domain that is composed of a repeating amino acid sequence with glycine, the smallest of the amino acids, in every third position and primarily separated by proline residues (Yamada et al. 1980; Bernard et al. 1983; Exposito et al. 2002; Boot-Handford and Tuckwell 2003; Aouacheria et al. 2004; Wada et al. 2006). This repetition is crucial, as it enables the triple-helix domain to wind into its compact structure in type I collagen. As such, the most phenotypically severe (e.g., OI type II) DAMs often result from substitutions of these glycines, whereas less severe phenotypes (e.g., OI type I) often result from alterations of the length of the triple-helix domain, which is normally encoded by 43 of 51 exons (fig. 1) within the >17-kb *COL1a1* locus (Dalgleish 1997; Kuivaniemi et al. 1997; Marini et al. 2007; Rauch et al. 2010). Studies have attempted to predict the phenotypic severity associated with COL1a1 mutations based on their amino acid position and affect on the thermostability of type I collagen (e.g., Persikov et al. 2005; Marini et al. 2007; Bodian et al. 2008, 2009); however, surprisingly, no

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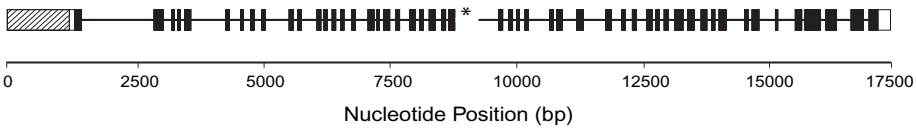

Fig. 1. The *COL1a1* gene locus, with all coding and noncoding regions to scale. DNA sequenced region in human and chimpanzee includes 1,223 bp of the promoter (striped box), 263 bp of the 5′ and 3′ mRNA untranslated regions (white boxes), the 4,392 bp of coding sequence for all 51 exons (black boxes), with noncoding regions in between. The "*" denotes a 496-bp gap in the collected sequence of intron 25 due to the presence of an *Alu* element.

model has incorporated an evolutionary approach. For example, genome-wide studies (e.g., Miller and Kumar 2001; Subramanian and Kumar 2006) using evolutionary site models have shown that DAMs in general are found more often at amino acid positions that are highly conserved across species.

Because previous efforts to identify DAMs at *COL1a1* have focused primarily on coding regions, it is unclear whether variation in other gene regions is functionally relevant. In fact, given the abundance of rare disease-associated amino acid variants (Dalgleish 1997; Marini et al. 2007) and the rarity of COL1a1 protein variation in general in the natural population (Chan et al. 2008), the observed variation in bone strength among human populations implicates *COL1a1* noncoding regions as functionally relevant as well. For example, an Sp1 transcription factor binding site mutation in the first intron of *COL1a1* increases gene expression and likely accounts for the associated change in the ratio of COL1a1 and COL1a2 subunits that reduces the structural integrity of type I collagen (Grant et al. 1996; Mann et al. 2001; Jin et al. 2009). Interestingly, this mutation reaches frequencies >20% in certain populations, but is absent in others, thus contributing to ethnic and geographic variation in bone strength and fracture risk (Bandres et al. 2005; Ralston et al. 2006; Jiang et al. 2007).

Potentially, functional noncoding variation associated with gene expression may also include intron compositional properties that impact gene structure. For example, genome-wide studies have shown that highly expressed genes have higher GC content (Urrutia and Hurst 2001; Comeron 2004; Kudla et al. 2006), as well as shorter introns, which is related to transcriptional efficiency (Hurst et al. 1996; Castillo-Davis et al. 2002; Urrutia and Hurst 2003; Comeron 2004). *COL1a1* is likely highly expressed given its abundance in connective tissue and its importance during fetal development and wound repair (e.g., Gelse et al. 2003; Hildebrand et al. 2005; Cohen 2006), yet because studies are often constrained to analyses of soft tissues (e.g., Su et al. 2004; Blekhman, Oshlack, et al. 2008), little is known about *COL1a1* gene expression. In this respect, in addition to evolutionary analyses of *COL1a1* coding regions, similar analyses of introns (e.g., length and GC content) could also reveal historical selective pressures related to functional constraint.

Attempts have been made to predict mutations associated with bone-related disease, the majority of which include COL1a1 because of its direct link to skeletal phenotypes. However, these studies have focused on protein sequences and, more specifically, have often ignored an evolutionary approach. Here, we present the first molecular evolutionary and statistical analysis of both *COL1a1* coding and noncoding sites to ask several questions: 1) Is there evidence of differential selective pressure across protein domains and how does this vary across recent/historical periods of time? 2) Do amino acid sites of high and low functional constraint over evolutionary time predict the gene locations and severity of human DAMs? 3) Finally, are there particular aspects of the gene structure that are evolutionarily unique compared with even other fibrillar collagens?

Materials and Methods

DNA Amplification and Sequencing

Our DNA sequences were collected either from public genome databases (National Center for Biotechnology Information (NCBI) and Ensembl) or generated directly when necessary. For example, with an estimated molecular divergence time from our common ancestor at ~4–6 My (e.g., Kumar et al. 2005), the chimpanzee is our closest living primate relative and is necessary for estimates of *COL1a1* nucleotide site divergence in the human lineage. However, regions of the chimpanzee (and to some extent the human) *COL1a1* gene sequences have assembly errors in their genome databases (which is likely a result of the repetitive nature of the sequence). Thus, DNA sequences for the >17-kb chromosome 17q21.33 locus (fig. 1) were generated from a human and a western Africa *Pan troglodytes verus* chimpanzee, both sampled from the Coriell Institute for Medical Research (Camden, NJ). Although there are several recognized chimpanzee subspecies, *P. t. verus* is the most appropriate contrast with humans because it has similar levels of nuclear diversity (Stone et al. 2002; Gilad et al. 2003; Fischer et al. 2004; Wooding et al. 2005; Verrelli et al. 2006, 2008; Claw et al. 2010).

The overall high GC content at *COL1a1* (discussed below) often required that our polymerase chain reaction (PCR) fragments are generated in short sequences (e.g., as small as 500 bp) with various temperature and buffer conditions. PCR products were purified using shrimp alkaline phosphatase and exonuclease I (US Biochemicals, Cleveland, OH) prior to DNA sequencing using an ABI 3730 automated sequencer (Applied Biosystems, Foster City, CA). Sequences were aligned and edited using Sequencher v. 4.5 (Gene Codes, Ann Arbor, MI). All PCR and sequencing primers were designed from available

human sequence (accession no. NT_010783.14) and are available upon request.

Gene Structure Analyses

To test hypotheses about selective pressures acting on gene structure, we first used a gene family approach comparing *COL1a1* in humans with other closely related fibrillar collagens. Phylogenetic comparisons of vertebrate fibrillar collagen proteins reveal three clades with *COL1a1* being one of five "clade A" collagens (Boot-Handford and Tuckwell 2003; Aouacheria et al. 2004; Wada et al. 2006). As such, length and GC content data for all exons and introns were collected from public databases for the four other clade A collagens (*COL1a2*, *COL2a1*, *COL3a1*, and *COL5a2*). Sequence gaps and *Alu* elements were omitted, the latter of which to avoid polymorphic insertions (e.g., a polymorphic *Alu* in *COL3a1*; Milewicz et al. 1996). Furthermore, estimates of intron nucleotide divergence were calculated from sites that best reflect "neutrality." Although we cannot say with certainty which sites are selectively neutral, we do expect that purifying selection acts relatively stronger on certain sites with putative function compared with others (e.g., McDonald and Kreitman 1991; Wray et al. 2003). Thus, we omitted first introns, which are typically enriched for transcription factor binding sites (e.g., Bornstein et al. 1987; Majewski and Ott 2002), as well as intron 5′ and 3′ splice sites. Whereas this first comparison examines the "uniqueness" of *COL1a1* among fibrillar collagens, a second comparison was used to also gain an overall genomic perspective. We used the Gazave et al. (2007) data set, which includes length, GC content, and human–chimpanzee nucleotide divergence estimates for a total of 51,673 introns compiled from 7,791 genes, representing the largest evolutionary analysis of human genome introns. As with above, first introns and intron splice sites were omitted.

Because we are interested in whether specific aspects of *COL1a1* introns are unusual, both independently and with respect to overall structure, analyses of "means" alone for intron length and GC content are not informative. Furthermore, the distribution of intron length in the human genome is dramatically skewed (i.e., lacks "normality," Gazave et al. 2007); thus, because of the enormity of the genome-wide sample relative to *COL1a1*, even nonparametric tests that compare only means of distributions may not be statistically sensitive. Instead, we constructed bins for the distributions of intron "length" and of GC content "percentage" for each clade A collagen and for the Gazave et al. (2007) data set and compared these binned distributions using nonparametric row by column (R × C) chi-squared tests (e.g., supplementary fig. 1, Supplementary Material online). Among clade A collagens, comparisons of binned distributions were also made for exon length and GC content. These analyses that compare distributions of site frequency classes allow for fine-scale statistical tests and increased power, especially in cases of unequal sample size (e.g., Akashi and Schaeffer 1997; Verrelli and Tishkoff 2004; Hernandez et al. 2007). Statistical significance levels for all tests were corrected for multiple comparisons using a standard Bonferroni method.

We also carried out comparative analyses across taxa to estimate how aspects of the gene structure of *COL1a1* may have changed over evolutionary time. To avoid errors in alignment and problems associated with incomplete sequences from available databases, we only used species for which complete *COL1a1* intron and exon sequences were publicly available. That is, species with gaps in their genomic sequence, regardless of the length of the gap, were excluded. As a result, the only species for which complete, high-quality *COL1a1* sequences were publicly available are mouse (*Mus musculus*), dog (*Canis familiaris*), cow (*Bos taurus*), western-clawed frog (*Xenopus tropicalis*), and zebrafish (*Danio rerio*), which were added to our human and chimpanzee sequences. This "seven-species data set" reflects a broad sampling across ~450 My of vertebrate evolution. Unlike our previous comparisons within the human genome, these analyses involve the same number of exons and introns. Thus, comparisons across species were conducted for *COL1a1* exon and intron length and GC content using Mann–Whitney *U* tests. To test hypotheses regarding the conservation of independent introns across evolutionary time, we also used nonparametric R × C chi-squared tests to compare orthologous introns and exons among these species (e.g., supplementary fig. 2, Supplementary Material online).

Nucleotide Site Divergence

To estimate the impacts of long- and short-term selective pressures on the COL1a1 protein, we first constructed a "vertebrate" data set reflecting ~450 My from available complete cDNA sequences, which includes the seven-species data set in addition to rat (*Rattus norvegicus*), donkey (*Equus asinus*), African clawed frog (*Xenopus laevis*), rainbow trout (*Oncorhynchus mykiss*), and goldfish (*Carassius auratus*). Amino acid sequences (which were the only available complete sequence) from chicken (*Gallus gallus*) and Japanese firebelly newt (*Cynops pyrrhogaster*) were also collected. Second, we also constructed a "primate" data set using our human and chimpanzee sequences in combination with database partial cDNA sequences (see supplementary fig. 3, Supplementary Material online) from gorilla (*Gorilla gorilla*), orangutan (*Pongo abelii*), macaque (*Macaca mulatta*), and marmoset (*Callithrix jacchus*), reflecting ~45 My in total. Sequences within the two data sets were aligned with a Clustal analysis in MEGA v.4 (Kumar et al. 2008), with the removal of alignment gaps.

With these two data sets, we conducted several analyses of the ratio of divergence at nonsynonymous and synonymous sites (*dN*/*dS*; e.g., Nei and Gojobori 1986). To identify statistically significant differences, we used the GABranch algorithm of Pond and Frost (2005b) available through the Datamonkey on-line server (Pond and Frost 2005a). This analysis enables more flexible testing of hypotheses in identifying different *dN*/*dS* evolutionary

Table 1. Human Clade A Collagen Gene Exon and Intron Characteristics.

	Gene				
Characteristics	*COL1a1*	*COL1a2*	*COL2a1*	*COL3a1*	*COL5a2*
Gene size (bp)	16,012	35,362	30,915	37,285	145,535
No. of exons	51	52	54	51	54
Exon length (bp)	86 ± 52	79 ± 49	83 ± 53	86 ± 53	83 ± 54
Intron length (bp)	193 ± 136	563 ± 502	416 ± 442	454 ± 316	1,380 ± 1,297
Exon GC content (%)	66 ± 5	59 ± 8	64 ± 6	59 ± 7	57 ± 7
Intron GC content (%)	59 ± 7	35 ± 5	55 ± 6	30 ± 4	32 ± 4

NOTE.—Gene size is the distance from the start to the stop codon. Length and GC content values denote means and standard deviations, excluding first introns and splice sites. See supplementary figs. 1 and 7, Supplementary Material online for more information.

"classes" of constraint using a maximum-likelihood approach. For both our general vertebrate and primate data sets, the relationships among species were described with a maximum parsimony tree constructed in MEGA v.4 (Kumar et al. 2008), and the best-fit nucleotide substitution model for each data set was determined by the selection procedure of Pond and Frost (2005c).

Other than the triple-helix domain, COL1a1 contains N- and C-terminal noncollagenous domains, which are found in all fibrillar collagens and hypothesized to have different functional constraints (e.g., Exposito et al. 2002; Boot-Handford and Tuckwell 2003; Aouacheria et al. 2004). Thus, we also applied the GABranch algorithm (Pond and Frost 2005b) to each COL1a1 domain separately and used the HyPhy program (Pond et al. 2005) to assess statistical significance. Specifically, likelihoods are generated from a substitution rate model for *dN/dS* evolving independently across all lineages in each domain and then compared with a second set of likelihoods (using a standard likelihood ratio test employed by HyPhy) generated from models where each domain is constrained to fit another. For example, is the rate for the triple-helix different from that observed in the N- or C-terminal across lineages?

Unlike the coding region analyses above, noncoding regions are sufficiently diverged to preclude alignment across taxa, even for closely related nonhuman primates in our data set. Thus, we estimated historical selective constraint on the *COL1a1* intron nucleotide sequence in humans with an analysis of human–chimpanzee divergence. As with above, we wish to examine how divergence varies across introns (which is especially relevant to our analysis of intron lengths) and not compare simple "means"; thus, we used similar R × C chi-squared tests as above. Specifically, after removing alignment gaps and splice sites, a distribution of divergence for *COL1a1* introns was constructed as bins of "percentage" (number of differences per nucleotide) and compared with similar bins of human–chimpanzee divergence for introns in the other four clade A collagen genes, as well as in the Gazave et al. (2007) data set.

DAMs at COL1a1

Finally, we examined the association of COL1a1 amino acid sites of high and low evolutionary conservation with where DAMs are located. Following the method of Subramanian and Kumar (2006), we estimated the evolutionary substitution rate at each amino acid site from our vertebrate data set (14 species listed above for which the entire amino acid sequence is available), using a maximum-likelihood approach implemented in PAML v.4.2 (Yang 2007). Specifically, the distribution of evolutionary rates among sites was categorized into an eight-class discrete gamma model (with an estimated gamma shape parameter starting at 0.5), and unequal substitution rates among amino acids were accounted for using the model of Jones et al. (1992). A total of 293 unique missense and nonsense COL1a1 DAMs that could be plotted within our vertebrate alignment were collected from the Database of Osteogenesis Imperfecta and Type III Collagen Mutations (Dalgleish 1997). We then used a chi-squared test to contrast the distribution of these observed mutations with the distribution of expected mutations in the eight-class model derived from the Subramanian and Kumar (2006) analysis.

As the majority of COL1a1 mutations are associated with OI, we also examined this distribution according to the clinical severity of disease symptoms (Sillence et al. 1979; Basel and Steiner 2009). Type I OI typically presents with mild bone weakening similar to that caused by osteoporosis; OI II, the most severe form, is typically lethal in the pre- or postnatal stage; OI III, the most severe of survivable OI types, is characterized by severe skeletal deformities and fragility fractures; and those that cannot be placed into types I–III are typically classified as OI IV. With this classification, we used four categories: 1) OI I and osteoporosis-associated mutations, 2) OI IV and related mutations (e.g., I/IV and III/IV), 3) OI III mutations, and 4) lethal OI II and II/III mutations, which totals 249 mutations that meet these criteria. We repeated the same chi-squared analyses above using the expected distributions from the Subramanian and Kumar (2006) analysis but with each of these four severity categories as the observed distributions. Finally, to determine if there is significant spatial clustering of these severity categories across the COL1a1 amino acid sequence, we used the maximum-likelihood approach of Zhang and Townsend (2009) as implemented in the MACML program.

Results

Exon Pattern Analyses

COL1a1 exons, while seemingly short in length, are not unusual for clade A fibrillar collagens (table 1; supplementary

Table 2. Clade A Collagen Gene Human–Chimpanzee Divergence Estimates.

	Gene									
	COL1a1		*COL1a2*		*COL2a1*		*COL3a1*		*COL5a2*	
Sites	*n*[a]	*d*[b]	*n*	*d*	*n*	*d*	*n*	*d*	*n*	*d*
Introns[c]	9,463	0.69	28,160	0.97	21,619	1.09	22,242	0.94	71,769	0.74
Synonymous	1,212	1.49	1,140	0.79	1,220	1.31	1,195	0.50	1,209	1.16
Nonsynonymous	3,180	0	2,958	0.10	3,240	0.03	3,203	0.03	3,288	0.06

[a] Number of nucleotides.
[b] Divergence as number of differences per nucleotide (%).
[c] Excludes first introns, see table 1 and supplementary fig. 7, Supplementary Material online for more information.

fig. 1, Supplementary Material online). Exon length does not differ between human and chimpanzee and although there are differences within our seven-species data set (which are primarily restricted to the N-terminal domain), they are not statistically unusual (supplementary fig. 2, Supplementary Material online). At 66%, *COL1a1* exon GC content is considerably higher than that seen at human genes in general (International Human Genome Sequencing Consortium 2001), which is not surprising given the abundance of GC-rich glycine and proline codons typically found in collagens. However, *COL1a1* exon GC content is still significantly greater than the other clade A fibrillar collagens (except *COL2a1*), even when examining only glycine- and proline-rich triple-helix coding regions (table 1; supplementary fig. 1, Supplementary Material online). Furthermore, although *COL1a1* exon GC content overall differs for comparisons between mammals and nonmammals, no differences are seen in comparisons of orthologous exons (supplementary fig. 2, Supplementary Material online).

Protein Evolution and DAMs

COL1a1 human–chimpanzee synonymous divergence is not unusual compared with other genes (e.g., Chimpanzee Sequencing Consortium 2005), whereas nonsynonymous divergence at *COL1a1* (as well as at all clade A collagens) is virtually absent (table 2). Low *COL1a1* amino acid divergence is also observed within our primate sample ($dN/dS < 0.15$, supplementary fig. 3, Supplementary Material online), and although the GABranch analysis found evidence for two dN/dS rate classes overall, the small number of fixations precluded analyses among domains. On the other hand, the analysis of our vertebrate sample found evidence for six dN/dS rate classes in the triple-helix domain ($dN/dS < 0.36$) and three rate classes for each of the C- and N-terminal domains ($dN/dS < 0.17$ and <0.74, respectively; supplementary fig. 4, Supplementary Material online). With the HyPhy analysis, we find little evidence for rate differences between the C- and N-terminal domains for comparisons between primate and other mammalian lineages, whereas the rate in the triple-helix is significantly less than that of the N-terminal but significantly greater than that of the C-terminal domain (likelihood ratio test, $P < 0.009$).

Using the approach of Subramanian and Kumar (2006), our site analysis finds spatial variation across COL1a1, with sites in the N-terminal domain having a significantly higher evolutionary rate (fig. 2). In addition, DAMs occur more often at highly conserved amino acid positions, a relationship that is consistent for all of the phenotypic severity categories associated with OI disease except, interestingly, the least severe category 1 (fig. 2; supplementary fig. 5, Supplementary Material online). In fact, when focusing on the extremes of lethal (category 4) and more benign, but osteoporetic-like (category 1) mutations, the former occur at much more highly conserved amino acid positions a significantly greater proportion of the time ($\chi^2 = 25.2$,

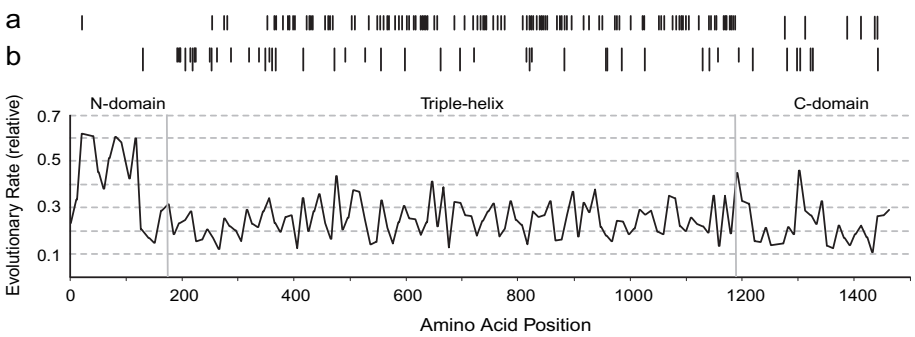

Fig. 2. Locations of DAMs are shown as vertical markers along the COL1a1 amino acid sequence for (*a*) category 4 mutations, which include lethal OI II and II/III groups and (*b*) category 1 mutations, which include OI I and osteoporosis-associated groups. Long and short vertical markers reflect nonglycine and glycine substitutions, respectively, across the three protein domains. The relative estimated evolutionary rates along the sequence are calculated from 14 diverse vertebrate taxa using the Subramanian and Kumar (2006) analysis, with a trendline summarizing a sliding window of averages across windows each of 10 amino acid residues (position number based on human sequence). See Materials and Methods and supplementary fig. 5, Supplementary Material online for information.

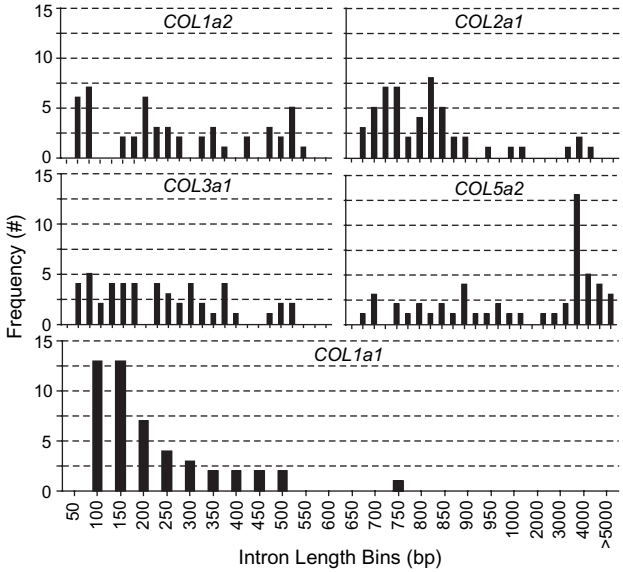

Fig. 3. Frequency distributions of intron length for human clade A collagen genes, excluding first introns. The *x* axis reflects the intron length bins (all scales are the same), and the *y* axis shows the number of introns within these bins. See supplementary fig. 7, Supplementary Material online for more information.

$P = 0.0001$). Finally, our MACML analysis finds clustering across COL1a1 in several regions, with statistically significant evidence involving the two extreme DAM categories, regardless of the model used from Zhang and Townsend (2009). A significant cluster for category 1 includes residues 190–366 ($P < 0.05$), which is the N-terminal end of the triple-helix domain, whereas a significant cluster for category 4 includes residues 352–1186 ($P < 0.05$), which is centered on the triple-helix domain.

Intron Pattern Analyses

COL1a1 introns are significantly shorter than those of the other clade A collagens with the exception of *COL2a1* (table 1 and fig. 3). In fact, the intron structure at *COL1a1*—many introns, but all of them relatively short in length—is statistically very unusual ($\chi^2 = 150$, $P < 10^{-6}$) when compared with human genes in general (Gazave et al. 2007). We do find evidence for significant variation in length of orthologous introns between taxa within our seven-species data set; however, as overall intron length has not changed dramatically (supplementary fig. 6, Supplementary Material online), *COL1a1* intron content appears to be similar even across vertebrates separated by ~450 My.

Intron GC content at *COL1a1* is significantly greater than that at the other clade A collagens except *COL2a1* (Supplementary fig. 7, Supplementary Material online). However, we also found that GC content was negatively correlated with intron length at *COL1a1* (supplementary fig. 7, Supplementary Material online), a pattern that has been generally noted from genome-wide studies (e.g., International Human Genome Sequencing Consortium 2001; Gazave et al. 2007; Pozzoli et al. 2008). Nonetheless, even when the data set for the five genes was standardized to include introns only 80–500 bp in length to reflect intron sizes at *COL1a1*, the results did not change. Unlike *COL1a1*, no other collagen gene shows a significant correlation between intron length and GC content; thus, instead of "percentage" GC content, we examined linear regressions of the number of G/C nucleotides in each intron, which standardizes GC content for intron length. *F*-test analyses comparing these regressions confirm that intron GC content at *COL1a1* is unusually higher than all other clade A collagens except *COL2a1* (supplementary fig. 7, Supplementary Material online). In fact, when compared with the Gazave et al. (2007) data set, and genes with introns of only 80–500 bp, *COL1a1* still has significantly greater GC content ($\chi^2 = 35$, $P = 0.0009$). Unlike intron length, intron GC content differs significantly in our seven-species data set at *COL1a1*, regardless of whether overall distributions or pairwise comparisons among equivalent introns are compared (supplementary fig. 6, Supplementary Material online). Overall, intron GC content is significantly higher in mammals versus nonmammals, except for mouse, which has significantly lower intron GC content than even other mammals.

Analyses of genome-wide introns indicate that length and GC content are also positively correlated with human–chimpanzee divergence (Gazave et al. 2007). Although this seems to be the case for intron length and divergence at *COL1a1* and the other clade A collagens, the relationship for GC content and divergence is only weakly correlated for the collagen genes (supplementary fig. 7, Supplementary Material online). Thus, as in analyses above, we standardized data sets to include introns 80–500 bp for divergence comparisons. Intron divergence at *COL1a1* is low but not statistically unusual compared with the other collagens (table 2; supplementary. fig. 7, Supplementary Material online) or human genes in general ($\chi^2 = 8.0$, $P = 0.6$). However, it is possible that our analysis of *COL1a1* divergence is an underestimate as it does not take into consideration the significantly high gene-specific GC content (e.g., Hernandez et al. 2007). To test this latter hypothesis, we used divergence at *COL1a1* non-CG dinucleotides (0.43%) as a conservative, "background" substitution rate (i.e., instead of the higher genome-wide estimate of ~1%; Chimpanzee Sequencing Consortium 2005). This estimate was adjusted for the ~10-fold increase in mutation rate expected for CG-dinucleotides in general (Subramanian and Kumar 2003) and compared with the rate "observed" at CG-dinucleotide sites (i.e., after using orangutan as an outgroup to correct for changes recently in the human and chimpanzee lineages). When these two estimates were applied to the total number of intron sites at *COL1a1* to simulate sampling variance, interestingly, observed human–chimpanzee divergence appears to be reduced ($\chi^2 = 6.14$, $P = 0.013$).

Discussion

Our evolutionary analyses of COL1a1 spanning the past ~450 My are consistent with it being one of the most

highly conserved in vertebrates. Although this may be the case for the protein overall, we find evidence for spatial/temporal variation in selective pressures across domains. In fact, the C-terminal domain that is responsible for the recognition and assembly of type I collagen subunits (e.g., Exposito et al. 2002; Boot-Handford and Tuckwell 2003; Aouacheria et al. 2004) is actually the most highly conserved domain. Like the N-terminal, the C-terminal peptide is cleaved after the subunit's triple-helix domain has been assembled into type I collagen; thus, interestingly, the protein region that exhibits the strongest signature of purifying selection is also one that is not part of the mature protein. On the other hand, the relatively less constraint on the COL1a1 triple-helix domain implies more "flexibility" in this region over evolutionary time, which may contribute to observed structural and mechanical variation in bone, including mineral content and organization of collagen fibers, among mammals, birds, and reptiles (e.g., Currey 1987; Wang et al. 1998; Rensberger and Watabe 2000).

It is possible that the evolutionary differences seen across domains are the result of directional selection pressures over time. Others have suggested this explanation (e.g., Morgan et al. 2010); however, the rate variation observed here is not what would be typically interpreted as a signature of "positive selection." Specifically, the estimates of *dN*/*dS* over both recent and deeper evolutionary lineages are more consistent with purifying selection, and thus, the differences among domains in these analyses, while statistically significant, are more reflective of variation in functional constraint and not adaptive evolution. Furthermore, our observations of very little amino acid variation among primates, including no fixation between human and chimpanzee, together with the observed slight differences in the properties of bone documented among primates (e.g., Wang et al. 1998; Kikuchi et al. 2003), are consistent with strong purifying selection that simply varies from ancient to more recent evolutionary time.

The abundance of DAMs at COL1a1 is also consistent with high functional constraint; however, the fact that so many exist and yet are not associated with lethality (e.g., category 1 mutations) may suggest that purifying selection has recently become weak in the human lineage. This hypothesis has been championed as a general explanation for human disease prevalence as high amino acid polymorphism in humans (albeit rare in frequency), in spite of the little amino acid divergence with chimpanzees, is a genome-wide pattern (e.g., Bustamante et al. 2005; Boyko et al. 2008; Lohmueller et al. 2008). However, the fact that purifying selection, and the absence of any positive selection, dominates the evolutionary history at COL1a1 is somewhat novel compared with other genes strongly associated with human disease (e.g., Blekhman, Man, et al. 2008). Thus, instead of invoking weak purifying selection in the human lineage, a better explanation for the pattern of DAMs seen today is that selective constraints simply differ across protein domains, which is also consistent with the overall vertebrate pattern at COL1a1.

In fact, our comparisons of the location of DAMs with the estimates of evolutionary constraints at their respective sites help further support this hypothesis. For example, although previous studies have suggested that the probability of lethality for a mutation increases as one moves away from the N-terminus (e.g., Byers et al. 1991; Marini et al. 2007; Rauch et al. 2010), we find that DAMs not only cluster in specific protein sites of high evolutionary constraint but that they even cluster with respect to classes of lesser severity, including more common, osteoporetic-like conditions. Interestingly, the N-terminal domain, where only four DAMs have been observed (Dalgleish 1997), is also the region here that shows the least evolutionary constraint over recent as well as deep evolutionary time. On the other hand, ~7% of COL1a1 positions, including those in the triple-helix, lack known DAMs and are also the most rapidly evolving sites. Thus, while factors such as amino acid size, thermostability, and domain are important to bone-related disease prediction models (e.g., Persikov et al. 2005; Marini et al. 2007; Bodian et al. 2008, 2009), an evolutionary site model may detect bone-related phenotypes of lesser, but more common, severity with finer resolution, particularly for nonglycine mutations, which have been largely ignored by previous prediction models.

Our final analyses of the evolutionary history of the overall gene structure also find unusual patterns for noncoding DNA. Of particular interest is that *COL1a1* introns have remained relatively short over the past ~450 My despite divergence in overall genome size (e.g., haploid human genome is ~2-fold larger than that of zebrafish; Hedges and Kumar 2002). Nonetheless, there are several reasons why these patterns are not expected a priori. First, because of the repetitive nature of vertebrate fibrillar collagen coding sequence, which has been proposed to have originated through a series of duplication events from an ancestral collagen with a single exon of ~54 bp (Bernard et al. 1983; Valkkila et al. 2001; Exposito et al. 2002; Boot-Handford and Tuckwell 2003; Aouacheria et al. 2004; Wada et al. 2006), there is extensive homology among gene regions. This homology presents the potential for increased unequal crossing-over events that result in exon duplications and deletions in the triple-helix domain, which, although important in the history of collagen evolution, is highly deleterious today as it typically leads to severe disease (e.g., Barsh et al. 1985; Cohn et al. 1993; Bodian et al. 2009). As such, separating the short *COL1a1* exons to reduce "interexon homology" (e.g., Cohn et al. 1993) would actually predict longer introns, yet this is not observed here.

A second consideration involves the greater mutation rate associated with high GC content in general (e.g., Subramanian and Kumar 2003), and thus, the expectation that mutational input at *COL1a1*, including introns, may be relatively high. However, our conservative estimates suggest that intron site divergence at *COL1a1* may actually be reduced. We may expect that purifying selection on deleterious amino acid variants coincidentally reduces neutral linked polymorphism in introns (as a result of background

selection, e.g., Charlesworth et al. 1995). However, under this scenario, we do not expect intron site divergence, if neutral, to be reduced as gene regions become effectively unlinked and evolutionarily independent over longer periods of time (e.g., Hellmann et al. 2003). For example, even at *COL1a1*, synonymous sites show typical levels of neutral divergence, yet nonsynonymous sites are highly conserved. Thus, the pattern found for intron site divergence at *COL1a1* may be consistent with purifying selection, which suggests that intron site variation has functional implications as well.

As previously noted, transcriptional efficiency is correlated with GC content and intron length across the genome (Hurst et al. 1996; Castillo-Davis et al. 2002; Urrutia and Hurst 2003; Vinogradov 2003; Comeron 2004; Kudla et al. 2006; Pozzoli et al. 2008); thus, one explanation for the significantly short GC-rich introns at *COL1a1* is that high gene expression is maintained by stabilizing selection. As type I collagen is the most abundant protein in mammals and is consistently required across life stages, from fetal development to wound repair (e.g., Gelse et al. 2003; Hildebrand et al. 2005; Cohen 2006), strong selection for *COL1a1* transcriptional efficiency would appear to be necessary. In addition, for a gene the size of *COL1a1*, the observation of so few transposable elements (four, including one *Alu*) is unusual given their frequency in the genome, yet interestingly, this pattern is also consistent with human genes with high expression (International Human Genome Sequencing Consortium 2001; Hackenberg et al. 2005; Pozzoli et al. 2008). *COL2a1* may have similarly short GC-rich introns with few transposable elements (4, again including only 1 *Alu*) as *COL1a1* because of similar constraints on gene expression. This may be the case as type II collagen, encoded by *COL2a1*, is the main structural protein of cartilage and is, therefore, also as functionally important in bone development (Gelse et al. 2003; Cohen 2006). Finally, while certain introns (e.g., first introns) may exhibit lower evolutionary divergence as they are known to be enriched for transcription factor binding domains, reduced divergence across *COL1a1* introns overall may also be related to intron length. For example, transcription enhancers often accumulate within the ~150 bp of exon–intron boundaries (e.g., Majewski and Ott 2002), and thus, given the relatively short intron lengths at *COL1a1*, purifying selection may be expected to be higher if the number of intron nucleotide sites that are selectively neutral is also relatively reduced.

Conclusion

Although previous analyses of the COL1a1 protein have invoked positive selection to explain the molecular rate heterogeneity observed across both lineages and time, our data set including ~450 My of vertebrate evolution concludes that these patterns are best explained by variation in purifying selection. Furthermore, although low evolutionary *COL1a1* site rates are consistent with the abundance of DAMs found in the human population today, our unique approach may predict not only the location of these mutations but also their degree of severity. In fact, these patterns imply that COL1a1-related diseases are likely not isolated to humans, or primates in general, which suggests that even distantly related vertebrates would be suitable models for research on common bone-related diseases such as osteoporosis. Finally, the unusual patterns seen for the *COL1a1* intron structure may be a signature of selective constraint for high gene expression that has interesting implications. Specifically, given the inferred history of purifying selection on them, these noncoding regions should also be considered when predicting which variants at *COL1a1* may explain potentially deleterious, or even adaptive, bone-related phenotypes today.

Supplementary Material

Supplementary figures 1–7 are available at *Molecular Biology and Evolution* online (http://www.mbe.oxfordjournals.org/).

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